

```
from google.colab import files
uploaded = files.upload()
```

Screenshot ... 143144.png

Screenshot 2026-07-20 143144.png(image/png) - 133941 bytes, last modified: 20/07/2026 - 100% done
Saving Screenshot 2026-07-20 143144.png to Screenshot 2026-07-20 143144 (2).png

```
import cv2
import matplotlib.pyplot as plt

filename = list(uploaded.keys())[0]
img = cv2.imread(filename, 0)

sobel_x = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)

sobel_xy = cv2.magnitude(sobel_x, sobel_y)

plt.imshow(sobel_xy, cmap='gray')
plt.title('Sobel XY Edge Detection')
plt.axis('off')
plt.show()
```

Sobel XY Edge Detection

